



Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

College of Computer Studies

ITEC 204 – Data Structures and Algorithms

Name: Ma. Loraine P. Alminario	Date: Sept. 25, 2026
Section: BSIT - 2B	Instructor: Ms. Maxyne Nuela Ignacio

- I. **Objective:** Write a short explanation of what your program is designed to do. (Minimum of 3 sentences)

Answer: In an IT support setting, the program is intended to handle incident tickets produced by automation bots. A straightforward menu-driven system enables the user to add, display, search, remove, and count active incident tickets. The program illustrates fundamental linear data structure operations and arranges the incident records using a singly linked list.

- II. **Program Operations:** Briefly explain how your program performs each operation. (Insert Specific Code Line and Your Explanation)

Add Incident:

Specific Code Line: `new_ticket = Ticket(incident_id, bot, description)`

Explanation:

- This code creates a new ticket node containing the Incident ID, Bot name, and Short Description. The new ticket is then added to the end of the singly linked list.

Display Incidents:

Specific Code Line:

```
While current is not None:
    print(
        f'{current.incident_id:<15}'
        f'{current.bot:<20}'
        f'{current.description}'
    )
    Current = current.next
```

Explanation:

- This code goes through each node in the linked list and displays the Incident ID, Bot, and Short Description. The `current = current.next` statement moves to the next ticket until all active incidents are displayed.

Search Incident:

Specific Code Line: `if current.incident_id.lower() == incident_id.lower():`

Explanation:



Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

College of Computer Studies

- This code compares the Incident ID entered by the user with the Incident IDs stored in the linked list. If a matching ID is found, the program displays the details of the incident ticket.

Remove Resolved Incident:

Specific Code Line: `Current.next = current.next.next`

Explanation:

- This code removes a resolved ticket from the linked list by changing the connection between nodes. The ticket is skipped from the list, allowing the remaining active tickets to stay connected.

Count Active Incidents:

Specific Code Line: `Count += 1`

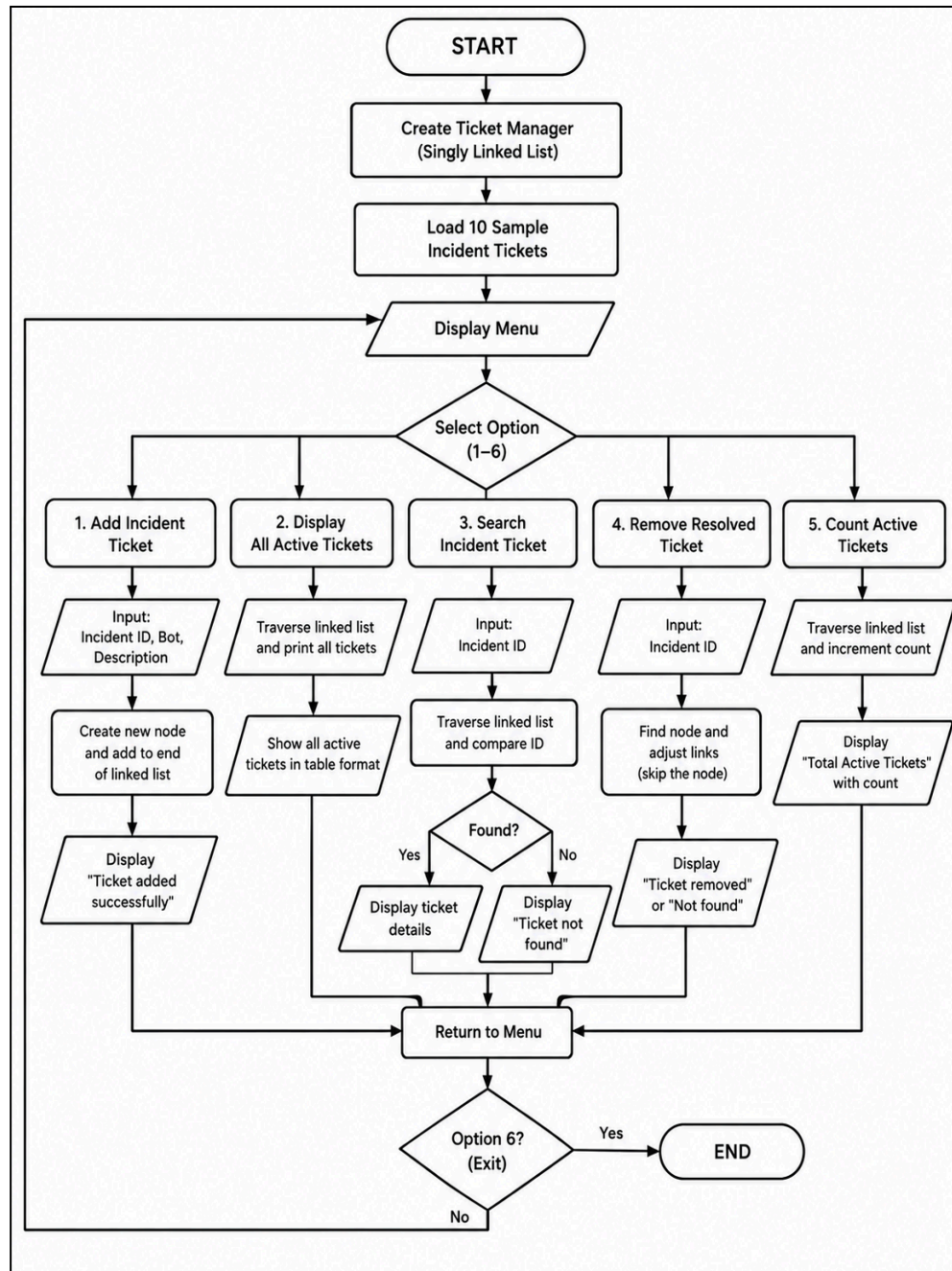
Explanation:

- This code increases the counter by one for every active incident ticket found in the linked list. After going through all the nodes, the program displays the total number of active tickets.



College of Computer Studies

III. **Flowchart:** Create a flowchart showing the overall process and logic of your program.



IV. **Conclusion:** Briefly explain what you learned from the activity and how the linear data structure was applied.

- This exercise taught me how to manage records in a real-world IT scenario using a linear data structure. I discovered how to use a singly linked list to add, display, search, remove, and count incident tickets. My understanding of how nodes are connected by pointers and how altering these connections can add or remove records from the list was also improved by the exercise.